

CAMBRIDGE INTERNATIONAL AS & A LEVEL CHEMISTRY (9701)

Topic 14 — Hydrocarbons

Topical Past Paper Question Pack & Comprehensive Mark Scheme

CANDIDATE	Urwah
SYLLABUS	Cambridge International AS Level Chemistry (9701)
TOPIC COVERAGE	14.1 Alkanes (Cracking, Combustion, Free Radicals) · 14.2 Alkenes (Electrophilic Addition, Oxidation, Polymerisation)
EXAM SERIES	Past Paper Compilation (2019 – 2025 Paper 2 Theory & Paper 1)
TOTAL MARKS	158 Marks (50 Questions)

HOW TO USE THIS PACK

1. Complete each question carefully in the answer spaces provided, paying close attention to mark allocations.
2. Questions are classified into Easy and Hard tiers to guide progression from core recall to advanced data analysis.
3. Every question includes authentic Cambridge examination codes (e.g. 9701/22/M/J/22) for cross-referencing.
4. Full examiner marking points with point-by-point breakdowns are provided at the end of the document.

SYLLABUS SUBTOPICS COVERED

14.1 Alkanes

— Crude oil refining, catalytic cracking (zeolite), complete/incomplete combustion (CO hazard), and

14.2 Alkenes— Electrophilic addition reactions (H₂, Br₂, HX, steam), Markovnikov's rule and carbocation stability, and

SUBTOPIC 14.1 — ALKANES: CRACKING, COMBUSTION & FREE-RADICAL SUBSTITUTION

1 Mechanism of Free-Radical Substitution of Methane — 9701/21/M/J/23/Q6(a)–(d)

HARD · Syllabus 14.1

Methane reacts with chlorine gas in the presence of ultraviolet light to yield chloromethane and hydrogen chloride. Fig. 1.1 outlines the three stages of the reaction mechanism.

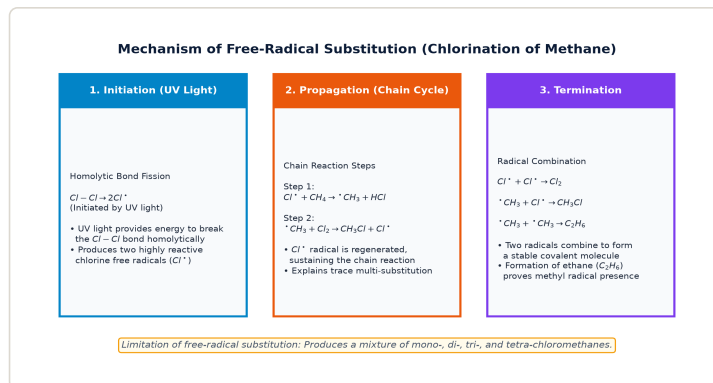


Fig. 1.1: Free-radical substitution mechanism for the chlorination of methane.

- ((a)) Write the equation for the initiation step, name the type of bond fission that occurs, and state the essential reaction condition. [3]

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- ((b)) Write the two equations that represent the propagation steps in the formation of chloromethane. [2]

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- ((c)) Write an equation for a termination step that accounts for the presence of trace amounts of ethane, C_2H_6 , in the product mixture. [1]

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- ((d)) State how the reaction conditions can be altered to maximise the yield of chloromethane, CH_3Cl , relative to polychlorinated products. [1]

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2 Catalytic vs Thermal Cracking of Alkanes — 9701/22/O/N/22/Q6(a)–(c)

EASY · Syllabus 14.1

Long-chain alkane molecules obtained from crude oil are cracked into more economically valuable shorter-chain alkanes and alkenes.

- ((a)) State the temperature and catalyst employed in industrial catalytic cracking. [2]

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- ((b)) Write a balanced equation for the cracking of decane, $\text{C}_{10}\text{H}_{22}$, to produce octane and one other product. [1]

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- ((c)) Explain why branched alkanes produced by catalytic cracking are preferred over straight-chain alkanes for use in motor car petrol. [1]
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3 Complete and Incomplete Combustion of Propane — 9701/23/M/J/22/Q5(a)–(c)

EASY · Syllabus 14.1

Propane, C₃H₈, is commonly supplied as liquefied petroleum gas (LPG) for heating and cooking.

- ((a)) Write a balanced equation for the complete combustion of propane. [1]
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- ((b)) Write a balanced equation for the incomplete combustion of propane forming carbon monoxide and water. [1]
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- ((c)) Explain why carbon monoxide is extremely hazardous to human health. [2]
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4 Isomeric Products from Bromination of Propane — 9701/21/O/N/21/Q6(a)–(c)

HARD · Syllabus 14.1

When propane reacts with bromine vapour in the presence of UV light, two monobromoalkane isomers are formed: 1-bromopropane and 2-bromopropane.

- ((a)) Write the structural formulae of both monobrominated isomers. [2]
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- ((b)) The major product is 2-bromopropane (approx 97% yield). Explain why 2-bromopropane is formed in much greater yield than 1-bromopropane, referring to radical intermediates. [3]
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5 Fractional Distillation of Crude Oil — 9701/11/M/J/22/Q23

EASY · Syllabus 14.1

In an industrial fractionating column, crude oil is separated into fractions.

- (i) Which property decreases as you move down the fractionating column from top to bottom? [1]
- A. Boiling point
 - B. Viscosity
 - C. Volatility
 - D. Flammability
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6 Environmental Hazards of Flue Soot Particulates — 9701/22/F/M/21/Q6(b)

EASY · Syllabus 14.1

Incomplete combustion of diesel fuel in compression engines produces carbon particulate matter (soot).

- (i)** State two adverse environmental or health effects associated with carbon particulates in the atmosphere. [2]

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7 Further Chlorination of Chloromethane — 9701/23/O/N/20/Q4(c)

HARD · Syllabus 14.1

When chloromethane reacts further with chlorine in the presence of UV light, dichloromethane is formed.

- ((a))** Write the two propagation steps for the conversion of chloromethane, CH_3Cl , to dichloromethane, CH_2Cl_2 . [2]

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- ((b))** State the molecular formula of the fully chlorinated end product formed when excess chlorine reacts with methane. [1]

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8 Thermal Cracking vs Catalytic Cracking Products — 9701/12/M/J/23/Q22

EASY · Syllabus 14.1

Which industrial cracking process is specifically optimised to produce high yields of alkenes such as ethene and propene?

- (i)** [1]

Select the cracking process:

- A. Catalytic cracking with zeolite at 500°C
- B. Thermal cracking at high temperature ($700\text{--}900^\circ\text{C}$) and high pressure (up to 70 atm)
- C. Steam reforming over nickel at 800°C
- D. Fractional distillation

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9 Combustion Stoichiometry Calculation — 9701/21/M/J/20/Q3(e)

HARD · Syllabus 14.1

A 20.0 cm^3 sample of a gaseous hydrocarbon C_xH_y was exploded with 150 cm^3 of oxygen (an excess). After cooling to room temperature, the total gas volume was 110 cm^3 . Passing this mixture through aqueous potassium hydroxide reduced the volume to 30 cm^3 (all gas volumes measured at RTP).

- ((a))** Calculate the volume of CO_2 produced in the explosion. [1]

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- ((b))** Calculate the volume of oxygen that reacted with the hydrocarbon. [2]

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- ((c))** Deduce the molecular formula of the hydrocarbon C_xH_y . [2]

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10 Unreactivity of Alkanes — 9701/11/O/N/22/Q21

EASY · Syllabus 14.1

Alkanes are generally unreactive towards polar reagents such as acids, alkalis, oxidising agents, and nucleophiles.

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Which factors explain the chemical inertness of alkanes under standard conditions?

- A. Low bond enthalpy of C-C bonds and low electronegativity of carbon.
- B. Non-polar nature of C-C and C-H bonds, and high bond enthalpies of C-C (347 kJ mol⁻¹) and C-H (413 kJ mol⁻¹).
- C. Presence of strong intermolecular hydrogen bonds.
- D. Planar geometry that shields carbon atoms from attack.

[1]

11 Relative Reactivity of Chlorine vs Bromine in Free-Radical Substitution — 9701/22/O/N/23/Q6(c)

HARD · Syllabus 14.1

Bromine is significantly more selective than chlorine in free-radical substitution reactions with alkanes.

((a))

Explain why the initiation step for bromine ($\text{Br}_2 \rightarrow 2\text{Br}^*$) requires less energy than the initiation step for chlorine ($\text{Cl}_2 \rightarrow 2\text{Cl}^*$).

[1]

((b))

Explain why the first propagation step with bromine is endothermic whereas with chlorine it is exothermic.

[2]

12 Identifying Termination Products — 9701/13/M/J/22/Q21

EASY · Syllabus 14.1

When ethane undergoes photochemical chlorination, which compound CANNOT be formed as a termination product?

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Select the impossible termination product:

- A. Butane, C_4H_{10}
- B. Chloroethane, $\text{C}_2\text{H}_5\text{Cl}$
- C. Chlorine, Cl_2
- D. Propane, C_3H_8

[1]

13 Greenhouse Impact of Methane vs Carbon Dioxide — 9701/21/M/J/21/Q5(a)

EASY · Syllabus 14.1

Methane, CH_4 , is a potent atmospheric greenhouse gas emitted by agricultural livestock and landfill sites.

((a))

Explain how greenhouse gases such as methane absorb infrared radiation and contribute to global warming.

[2]

((b))

Suggest why methane has a significantly higher global warming potential (GWP) per molecule than carbon dioxide.

[1]

14 Cracking Reaction Balancing — 9701/12/F/M/22/Q21

EASY · Syllabus 14.1

One mole of alkane X is cracked to produce one mole of octane, two moles of propene, and one mole of ethene.

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What is the molecular formula of alkane X?

- A. C₁₄H₃₀
- B. C₁₅H₃₂
- C. C₁₆H₃₄
- D. C₁₈H₃₈

[1]

SUBTOPIC 14.2 — ALKENES: ELECTROPHILIC ADDITION, OXIDATION & POLYMERISATION

15 Electrophilic Addition of HBr to Propene — 9701/22/M/J/23/Q6(a)–(d)

HARD · Syllabus 14.2

When hydrogen bromide reacts with propene, two isomeric bromoalkanes are formed. Fig. 15.1 details the two competing reaction pathways.

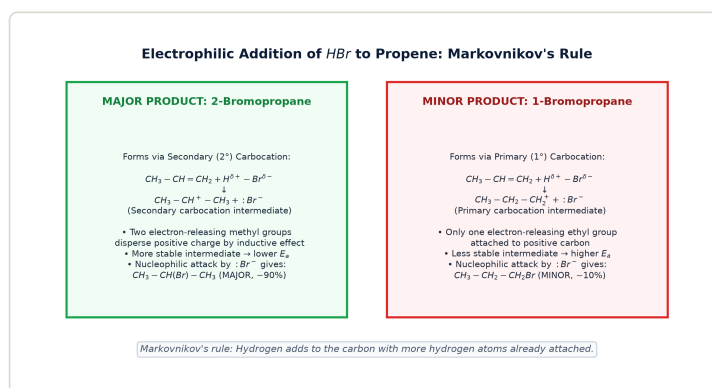


Fig. 15.1: Mechanism of electrophilic addition of HBr to propene.

((a))

Draw the complete mechanism for the reaction between propene and HBr leading to the major product. Include all partial charges, lone pairs, curly arrows, and intermediate charges.

[4]

((b))

State the systematic IUPAC name of the major product.

[1]

((c))

Explain why the major product is formed in much higher yield than the minor product, referring to the stability of the intermediate carbocations.

[3]

16

Test for Unsaturation: Bromine Water Addition — 9701/21/O/N/22/Q6(a)–(b)

EASY · Syllabus 14.2

Bromine water is the standard qualitative reagent used to distinguish alkenes from alkanes.

((a))

State the observation when bromine water is shaken with cyclohexene and with cyclohexane in the dark.

[2]

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((b))

Write the equation for the reaction of ethene with aqueous bromine to form 1,2-dibromoethane, and state the type of mechanism.

[2]

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17

Reaction of Ethene with Bromine Water: Bromohydrin Formation — 9701/23/M/J/21/Q5(c)

HARD · Syllabus 14.2

When ethene is bubbled through aqueous bromine containing sodium chloride, three organic products are formed: 1,2-dibromoethane, 2-bromoethanol, and 1-bromo-2-chloroethane.

((a))

Explain why 2-bromoethanol, $\text{CH}_2\text{BrCH}_2\text{OH}$, is formed as the predominant product in aqueous solution.

[2]

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((b))

Explain how 1-bromo-2-chloroethane is formed in this mixture.

[2]

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18

Industrial Catalytic Hydration of Ethene — 9701/22/F/M/22/Q6(a)–(c)

EASY · Syllabus 14.2

Ethanol is manufactured industrially by the direct catalytic hydration of ethene.

((a))

State the reagent and essential conditions (temperature, pressure, catalyst) for this process.

[3]

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((b))

Write a balanced chemical equation for the hydration reaction.

[1]

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((c))

State one environmental advantage of manufacturing ethanol by fermentation of sugar rather than by hydration of ethene.

[1]

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19 Catalytic Hydrogenation of Alkenes — 9701/11/M/J/23/Q23

EASY · Syllabus 14.2

Which catalyst and condition are commonly used for the industrial hydrogenation of vegetable oils to produce margarine?

(1)

Select the correct conditions:

- A. Nickel catalyst at 150 °C
 - B. Iron catalyst at 450 °C
 - C. Vanadium(V) oxide at 200 °C
 - D. Concentrated sulfuric acid at 100 °C
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[1]**20 Addition of Hydrogen Halides to 2-Methylbut-2-ene — 9701/21/M/J/22/Q6(b)**

HARD · Syllabus 14.2

Consider the addition of hydrogen chloride, HCl, to 2-methylbut-2-ene, $(\text{CH}_3)_2\text{C}=\text{CHCH}_3$.

((a))

Draw the skeletal formula and state the systematic IUPAC name of the major product.

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[2]**((b))**

Explain why 2-chloro-2-methylbutane is formed preferentially over 2-chloro-3-methylbutane.

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[2]**21 Induced Dipole in Halogen Addition Mechanisms — 9701/22/O/N/21/Q6(c)**

HARD · Syllabus 14.2

Although bromine, Br_2 , is a non-polar molecule, it acts as an electrophile when approaching an alkene.

((a))

Explain how a non-polar bromine molecule becomes polarised as it approaches the $\text{C}=\text{C}$ double bond.

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[2]**((b))**

State what happens to the $\text{Br}-\text{Br}$ bond during the first step of the electrophilic addition mechanism.

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[1]**22 Hydration of Propene: Major vs Minor Alcohol — 9701/12/O/N/22/Q22**

EASY · Syllabus 14.2

When propene is reacted with steam in the presence of an acid catalyst, which alcohol is formed as the major product?

(1)

Select the major alcohol product:

- A. Propan-1-ol
 - B. Propan-2-ol
 - C. Propanoic acid
 - D. Propanone
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[1]

23 Addition of Bromine to Buta-1,3-diene — 9701/23/O/N/21/Q5(d)

HARD · Syllabus 14.2

When one mole of bromine is added to one mole of buta-1,3-diene at low temperature, 3,4-dibromobut-1-ene is the kinetic product.

((a)) Write the structural formula of 3,4-dibromobut-1-ene. **[1]**

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((b)) At higher temperatures, the thermodynamic 1,4-addition product, 1,4-dibromobut-2-ene, predominates. Draw the skeletal formula of (E)-1,4-dibromobut-2-ene. **[2]**

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24 Reactions of Cycloalkenes with Acidic Reagents — 9701/13/O/N/21/Q21

EASY · Syllabus 14.2

What is the structural formula of the product formed when cyclohexene reacts with excess hydrogen bromide gas?

(i) Select the correct product: **[1]**

- A. 1,2-dibromocyclohexane
- B. Bromocyclohexane
- C. 1-bromocyclohexene
- D. 1,4-dibromocyclohexane

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25 Determining Alkene Structure from Hydrogenation Data — 9701/22/M/J/20/Q4(c)

HARD · Syllabus 14.2

A 0.100 mol sample of an unknown hydrocarbon requires 4.80 dm³ of hydrogen gas at RTP for complete saturation over a nickel catalyst. (Molar gas volume = 24.0 dm³ mol⁻¹).

((a)) Calculate the number of moles of C=C double bonds present per mole of hydrocarbon. **[2]**

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((b)) Suggest two possible structural formulas for this hydrocarbon if its molecular formula is C₄H₆. **[2]**

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26 Addition of Interhalogens to Alkenes — 9701/21/O/N/23/Q6(b)

HARD · Syllabus 14.2

Iodine monochloride, ICl, adds across the double bond of propene.

((a)) Deduce which atom in ICl acts as the electrophilic centre. **[1]**

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((b)) Predict the major organic product formed and state its systematic IUPAC name. **[2]**

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27 MCQ on Markovnikov's Rule — 9701/12/M/J/21/Q23

EASY · Syllabus 14.2

Which alkene produces ONLY ONE product upon reaction with hydrogen bromide?

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Select the symmetrical alkene:

- A. Propene
- B. But-1-ene
- C. But-2-ene
- D. 2-methylpropene

[1]

28 Oxidative Cleavage of Alkenes by Hot Concentrated KMnO_4 — 9701/21/M/J/23/Q6(e)-(g)

HARD · Syllabus 14.2

The oxidative cleavage of alkenes with hot concentrated acidified KMnO_4 is a powerful diagnostic tool for locating double bonds. Fig. 28.1 displays the three cleavage rules.

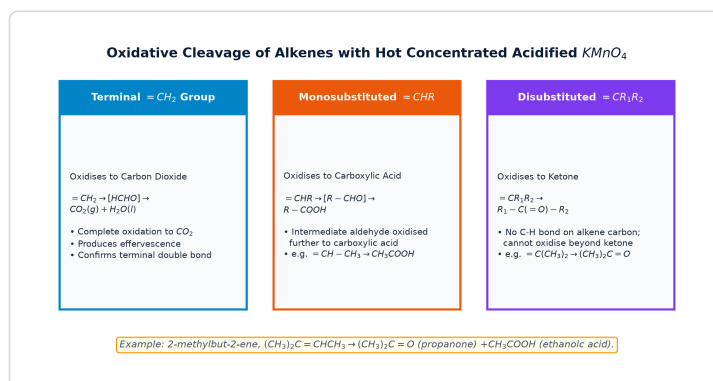


Fig. 28.1: Products obtained from oxidative cleavage of alkenes with hot concentrated KMnO_4 .

((a))

State the products formed when each of the following alkene fragments is cleaved by hot concentrated acidified KMnO_4 :

- (i) $=\text{CH}_2$
- (ii) $=\text{CH}-\text{R}$
- (iii) $=\text{C}(\text{R}_1)(\text{R}_2)$

[3]

((b))

An unknown hydrocarbon W, C_6H_{12} , is treated with hot concentrated acidified KMnO_4 . The only organic product obtained is propanoic acid, $\text{CH}_3\text{CH}_2\text{COOH}$. Deduce the displayed formula and systematic name of hydrocarbon W.

[2]

((c))

Alkene Y, C_5H_{10} , reacts with hot concentrated acidified KMnO_4 to produce propanone, $(\text{CH}_3)_2\text{C}=\text{O}$, carbon dioxide, and water. Deduce the structural formula and systematic name of alkene Y.

[2]

29 Mild Oxidation of Alkenes: Diol Formation — 9701/22/O/N/22/Q6(d)

EASY · Syllabus 14.2

When an alkene is shaken with cold, dilute, acidified potassium manganate(VII), a diol is formed.

((a)) State the observation during this reaction. **[1]**

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((b)) Write the balanced equation for the reaction of propene with cold dilute acidified KMnO_4 , representing the oxidising agent as $[\text{O}]$. **[2]**

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((c)) State the systematic IUPAC name of the diol product. **[1]**

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30 Deducing Alkene Structure from Cleavage Products — 9701/11/M/J/22/Q24

HARD · Syllabus 14.2

An alkene Z reacts with hot concentrated acidified KMnO_4 to produce butanone, $\text{CH}_3\text{COCH}_2\text{CH}_3$, and ethanoic acid, CH_3COOH .

(i) What is the structure of alkene Z? **[1]**

- A. 3-methylhex-2-ene
- B. 3-methylhex-3-ene
- C. 2-methylhex-2-ene
- D. 2-ethylpent-2-ene

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31 Oxidative Cleavage of Cyclic Alkenes — 9701/23/M/J/22/Q5(d)

HARD · Syllabus 14.2

When cyclohexene is refluxed with hot concentrated acidified potassium manganate(VII), the ring opens to form a single dicarboxylic acid.

((a)) Draw the structural formula of the dicarboxylic acid formed. **[1]**

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((b)) State the systematic IUPAC name of this dicarboxylic acid. **[1]**

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((c)) State how many moles of hot KMnO_4 are required per mole of cyclohexene. **[1]**

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32 Distinguishing Alkenes with Identical Formula — 9701/21/O/N/21/Q6(d)

HARD · Syllabus 14.2

Two unlabelled bottles contain liquid hex-1-ene and liquid hex-3-ene.

((a)) Explain why adding bromine water cannot distinguish between the contents of the two bottles. **[1]**

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((b)) Describe how hot concentrated acidified KMnO_4 can be used to distinguish between hex-1-ene and hex-3-ene, stating the expected observations. **[3]**

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33 Oxidation Products of 1,4-Cyclooctadiene — 9701/12/M/J/22/Q23

HARD · Syllabus 14.2

When 1-methylcyclopentene is cleaved by hot concentrated acidified potassium manganate(VII), which organic compound is formed?

0 **[1]**

Select the correct product:

- A. 5-oxohexanoic acid, $\text{CH}_3\text{COCH}_2\text{CH}_2\text{CH}_2\text{COOH}$
- B. Hexanedioic acid
- C. 6-hydroxyhexanoic acid
- D. Cyclopentanone and CO_2

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34 Infrared Monitoring of Alkene Oxidation — 9701/22/F/M/23/Q6(b)

HARD · Syllabus 14.2

An alkene is oxidised to a diol using cold dilute acidified KMnO_4 .

((a)) State the key absorption peak in the infrared spectrum that disappears upon complete conversion of the alkene to the diol. **[1]**

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((b)) State the prominent new absorption peak that appears in the product spectrum and identify the bond responsible. **[2]**

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35 Identifying Alkene Producing Carbon Dioxide — 9701/13/O/N/22/Q22

EASY · Syllabus 14.2

Which of the following alkenes will produce carbon dioxide gas when refluxed with hot concentrated acidified potassium manganate(VII)?

0 **[1]**

Select the alkene:

- A. But-2-ene
- B. 2-methylbut-2-ene
- C. 3-methylbut-1-ene
- D. Cyclopentene

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36 Oxidation of Terpenes (Limonene) — 9701/21/O/N/20/Q5(a)–(c)

HARD · Syllabus 14.2

Limonene is a natural hydrocarbon found in citrus fruit peels containing two C=C double bonds: an endocyclic $=C(CH_3)-CH=$ bond and an exocyclic $=C(CH_3)CH_2$ bond.

- ((a))** State the volume of bromine liquid (density 3.10 g cm^{-3}) needed to react completely with 0.100 mol of limonene. (Ar: Br = 79.9) **[3]**

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- ((b))** Explain why limonene yields carbon dioxide when refluxed with hot concentrated acidified $KMnO_4$. **[1]**

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37 Quantitative Gas Production from Alkene Cleavage — 9701/22/M/J/21/Q6(c)

HARD · Syllabus 14.2

A 0.0500 mol sample of an unknown alkene produces 1.20 dm^3 of carbon dioxide gas at RTP when treated with hot concentrated acidified $KMnO_4$. (Molar gas volume = $24.0\text{ dm}^3\text{ mol}^{-1}$).

- ((a))** Calculate the number of terminal $=CH_2$ groups per molecule of the alkene. **[2]**

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- ((b))** If the only other oxidation product is propanone, deduce the structure and name of the alkene. **[2]**

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38 MCQ on Cold $KMnO_4$ Oxidation — 9701/11/F/M/21/Q22

EASY · Syllabus 14.2

What is the organic product formed when but-2-ene is reacted with cold dilute acidified potassium manganate(VII)?

- 0** Select the correct product: **[1]**
- A. Ethanoic acid
 - B. Butane-2,3-diol
 - C. Butane-1,2-diol
 - D. Butan-2-one

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39 Comparative Reactivity: Alkane vs Alkene with $KMnO_4$ — 9701/23/M/J/20/Q4(b)

EASY · Syllabus 14.2

Hexane and hex-1-ene are tested with acidified $KMnO_4$ at room temperature.

- 0** State the observation for each compound and explain the difference in reactivity. **[3]**

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40 Addition Polymers: Monomers and Repeat Units — 9701/22/M/J/23/Q6(e)–(g)

HARD · Syllabus 14.2

Alkenes undergo addition polymerisation under suitable conditions of temperature, pressure, and catalyst. Fig. 40.1 illustrates common polymers and their repeat units.

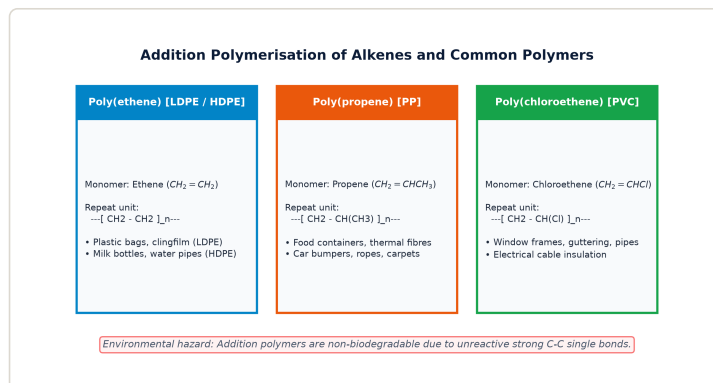


Fig. 40.1: Structures and repeat units of common commercial addition polymers.

((a)) Draw the displayed formula of the repeat unit of poly(propene). [2]

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((b)) A section of an addition polymer chain is shown below: [2]
 $\text{---CH}_2\text{---CH}(\text{Cl})\text{---CH}_2\text{---CH}(\text{Cl})\text{---CH}_2\text{---CH}(\text{Cl})\text{---}$
 Draw the displayed formula and state the systematic IUPAC name of the monomer used to synthesize this polymer.

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((c)) Explain why addition polymers such as poly(ethene) are chemically inert and non-biodegradable in landfill sites. [2]

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41 Environmental Problems with Plastic Waste Disposal — 9701/21/O/N/22/Q6(d)–(e)

HARD · Syllabus 14.2

The disposal of non-biodegradable synthetic polymers poses major environmental challenges.

((a)) State two environmental disadvantages of disposing of plastic waste in landfill sites. [2]

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((b)) Incineration is an alternative method of disposing of polymer waste. State one toxic pollutant gas released during the combustion of poly(chloroethene) (PVC), and describe how this gas can be removed from incinerator flue gases. [2]

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((c)) Explain what is meant by feedstock recycling of plastics. [1]

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42 Repeat Unit of Teflon (PTFE) — 9701/12/M/J/23/Q23

EASY • Syllabus 14.2

Poly(tetrafluoroethene) (PTFE / Teflon) is widely used for non-stick cookware coatings.

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What is the structure of the monomer used to manufacture PTFE?

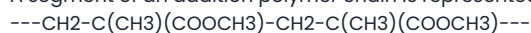
- A. $\text{CF}_3\text{-CF}_3$
- B. $\text{CF}_2=\text{CF}_2$
- C. $\text{CHF}=\text{CHF}$
- D. $\text{CF}_3\text{-CH}=\text{CF}_2$

[1]

43 Deducing Monomer from Polymer Section — 9701/23/M/J/22/Q5(e)

HARD • Syllabus 14.2

A segment of an addition polymer chain is represented below:



((a))

Draw the displayed formula of the monomer used to prepare this polymer.

[2]

((b))

State the systematic IUPAC name of the monomer.

[1]

((c))

State the trade name of this polymer widely used as an alternative to glass.

[1]

44 Addition Polymerisation of But-2-ene — 9701/11/O/N/21/Q22

HARD • Syllabus 14.2

Which structure represents the repeat unit of the polymer formed from but-2-ene?

0

Select the correct repeat unit:

- A. $\text{-[CH}_2\text{-CH}_2\text{-CH}_2\text{-CH}_2\text{]-}$
- B. $\text{-[CH(CH}_3\text{)-CH(CH}_3\text{)]-}$
- C. $\text{-[CH}_2\text{-CH(C}_2\text{H}_5\text{)]-}$
- D. $\text{-[C(CH}_3\text{)}_2\text{-CH}_2\text{]-}$

[1]

45 Co-Polymerisation Concept — 9701/21/M/J/21/Q6(b)

HARD • Syllabus 14.2

A copolymer is formed by the addition polymerisation of an equimolar mixture of ethene and propene.

((a))

Draw the structure of the repeating unit in an alternating copolymer of ethene and propene.

[2]

((b))

Calculate the mass of ethene required to react with 42.0 g of propene to form this alternating copolymer. (Mr: ethene = 28.0, propene = 42.0)

[2]

46 Biodegradable Polymers vs Addition Polymers — 9701/12/F/M/21/Q23

EASY · Syllabus 14.2

Why are condensation polymers (such as polyesters and polyamides) typically biodegradable, whereas polyalkenes are not?

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Select the correct explanation:

- A. Polyalkenes contain polar carbon-carbon bonds that repel water.
- B. Condensation polymers contain polar ester or amide linkages in their backbone that can be hydrolysed by water and bacterial enzymes.
- C. Condensation polymers have much lower molar masses than addition polymers.
- D. Addition polymers dissolve easily in rainwater.

[1]

47 Deducing Monomers from a Branched Polymer — 9701/22/O/N/20/Q5(d)

HARD · Syllabus 14.2

A synthetic rubber is an addition polymer of 2-methylbuta-1,3-diene (isoprene).

((a))

Draw the structural formula of 2-methylbuta-1,3-diene.

[1]

((b))

When 2-methylbuta-1,3-diene undergoes 1,4-addition polymerisation, a C=C double bond remains in each repeat unit. Draw the repeat unit of poly(isoprene).

[2]

((c))

Natural rubber is the cis-isomer of poly(isoprene), whereas gutta-percha is the trans-isomer. Explain why natural rubber is flexible and elastic while gutta-percha is rigid and inelastic.

[2]

48 Polymer Combustion and Energy Recovery — 9701/13/M/J/21/Q23

EASY · Syllabus 14.2

What is the primary advantage of incinerating polymer waste with energy recovery compared to landfilling?

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Select the main advantage:

- A. It produces no carbon dioxide emissions.
- B. The high calorific value of plastics can be harnessed to generate electricity or district heating, saving fossil fuels.
- C. It completely eliminates all atmospheric emissions.
- D. It generates reusable monomer liquids directly.

[1]

49 Recycling Codes and Mechanical Sorting — 9701/21/M/J/23/Q6(h)

EASY · Syllabus 14.2

Plastic containers are stamped with recycling identification codes (e.g. 1 for PET, 2 for HDPE, 3 for PVC, 4 for LDPE, 5 for PP, 6 for PS).

((a))

Explain why plastic waste must be sorted into specific polymer types before mechanical recycling.

[2]

((b)) State one physical method used in automated recycling facilities to separate different plastics. [1]

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50 Synoptic Problem: Hydrocarbon Reaction Pathways — 9701/21/O/N/23/Q6(a)–(f)

HARD · Syllabus 14.2

An unsaturated hydrocarbon A, C₄H₈, exists as a pair of stereoisomers.

- Hydrocarbon A decolourises bromine water to yield B.
- Reaction of A with hot concentrated acidified KMnO₄ yields a single organic product C, which effervesces with sodium carbonate.
- Compound A reacts with hydrogen gas in the presence of a nickel catalyst to form compound D.

((a)) Deduce the identity of hydrocarbon A and state the type of stereoisomerism it exhibits. [2]

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((b)) Draw the displayed formula of compound B and state its systematic IUPAC name. [2]

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((c)) Identify compound C and write the equation for its reaction with sodium carbonate. [2]

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((d)) Compound D is reacted with chlorine gas in the presence of UV light. State how many structural isomers of monochlorobutane are formed. [2]

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Mark Scheme — Topic 14 — Hydrocarbons

CAMBRIDGE INTERNATIONAL AS LEVEL CHEMISTRY (9701) — THEORY MARKING POINTS

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
1(a)	Equation: $\text{Cl}_2 \rightarrow 2\text{Cl}^*$ [1] Type of fission: Homolytic fission [1] Condition: Ultraviolet (UV) light / sunlight [1]	3
1(b)	Step 1: $\text{Cl}^* + \text{CH}_4 \rightarrow \text{CH}_3 + \text{HCl}$ [1] Step 2: $\text{CH}_3 + \text{Cl}_2 \rightarrow \text{CH}_3\text{Cl} + \text{Cl}^*$ [1]	2
1(c)	$\text{CH}_3 + \text{CH}_3 \rightarrow \text{C}_2\text{H}_6$ (or CH_3CH_3) [1]	1
1(d)	Use a large excess of methane (CH_4) relative to chlorine [1]	1
2(a)	Temperature: 450 – 550 °C (approx 500 °C) [1] Catalyst: Zeolite (or aluminosilicate / silica-alumina) [1]	2
2(b)	$\text{C}_{10}\text{H}_{22} \rightarrow \text{C}_8\text{H}_{18} + \text{C}_2\text{H}_4$ (octane and ethene) [1]	1
2(c)	Branched alkanes have a higher octane rating / burn more smoothly / reduce engine knocking [1]	1
3(a)	$\text{C}_3\text{H}_8 + 5\text{O}_2 \rightarrow 3\text{CO}_2 + 4\text{H}_2\text{O}$ [1]	1
3(b)	$\text{C}_3\text{H}_8 + 3.5 \text{O}_2 \rightarrow 3\text{CO} + 4\text{H}_2\text{O}$ (or $2\text{C}_3\text{H}_8 + 7\text{O}_2 \rightarrow 6\text{CO} + 8\text{H}_2\text{O}$) [1]	1
3(c)	Carbon monoxide binds irreversibly / with much higher affinity than oxygen to haemoglobin in red blood cells [1] forming carboxyhaemoglobin, preventing blood from transporting oxygen to body tissues, leading to asphyxiation [1]	2
4(a)	1-bromopropane: $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{Br}$ [1] 2-bromopropane: $\text{CH}_3\text{-CH(Br)-CH}_3$ [1]	2
4(b)	Formation of 2-bromopropane proceeds via the secondary propyl radical, $\text{CH}_3\text{-}\dot{\text{C}}\text{H-CH}_3$ [1] Formation of 1-bromopropane proceeds via the primary propyl radical, $\text{CH}_3\text{-CH}_2\text{-}\dot{\text{C}}\text{H}_2$ [1] The secondary radical is more stable than the primary radical due to the positive inductive electron donation from two methyl groups, so it forms with lower activation energy [1]	3
5	Correct Answer: C [1] Explanation: As you move down the column, molecules become larger with stronger London dispersion forces. Boiling point and viscosity increase, while volatility and flammability decrease.	1
6	Penetrate deep into human lungs, exacerbating asthma, bronchitis, and cardiovascular diseases [1] Deposit on buildings and statues causing blackening / contribute to global dimming by reflecting sunlight back into space [1]	2
7(a)	Step 1: $\text{Cl}^* + \text{CH}_3\text{Cl} \rightarrow \text{CH}_2\text{Cl} + \text{HCl}$ [1] Step 2: $\text{CH}_2\text{Cl} + \text{Cl}_2 \rightarrow \text{CH}_2\text{Cl}_2 + \text{Cl}^*$ [1]	2
7(b)	CCl_4 (tetrachloromethane) [1]	1
8	Correct Answer: B [1] Explanation: Thermal cracking at high temperature and pressure breaks C-C bonds near the chain ends via free radicals, maximising the yield of short-chain alkenes (feedstock for polymers).	1
9(a)	KOH absorbs CO_2 ; Volume of $\text{CO}_2 = 110 - 30 = 80 \text{ cm}^3$ [1]	1
9(b)	Unreacted $\text{O}_2 = 30 \text{ cm}^3$; Volume of O_2 reacted = $150 - 30 = 120 \text{ cm}^3$ [2]	2
9(c)	$20 \text{ cm}^3 \text{ C}_x\text{H}_y : 80 \text{ cm}^3 \text{ CO}_2 \Rightarrow 1 \text{ mol C}_x\text{H}_y : 4 \text{ mol CO}_2 \Rightarrow x = 4$ [1] Moles $\text{O}_2 = x + y/4 \Rightarrow 120/20 = 6 \Rightarrow 4 + y/4 = 6 \Rightarrow y/4 = 2 \Rightarrow y = 8 \Rightarrow \text{C}_4\text{H}_8$ (or butene/cyclobutane) [1]	2
10	Correct Answer: B [1] Explanation: Alkanes contain only non-polar sigma bonds with very small electronegativity differences ($\text{C}=2.5, \text{H}=2.1$), leaving no δ^+ or δ^- sites for nucleophiles/electrophiles to attack, and the covalent bonds have high bond enthalpies.	1
11(a)	The Br-Br bond enthalpy (193 kJ mol^{-1}) is lower than the Cl-Cl bond enthalpy (242 kJ mol^{-1}) [1]	1

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
11(b)	The H-Cl bond formed (431 kJ mol ⁻¹) is stronger than the C-H bond broken (413 kJ mol ⁻¹), making Delta-H negative (exothermic) [1] The H-Br bond formed (366 kJ mol ⁻¹) is weaker than the C-H bond broken (413 kJ mol ⁻¹), making Delta-H positive (endothermic) [1]	2
12	Correct Answer: D [1] Explanation: The radicals present during ethane chlorination are Cl* and *C2H5. Combinations produce Cl2, C2H5Cl, and butane (C2H5-C2H5). Propane requires a methyl radical (*CH3), which is not generated.	1
13(a)	Molecules absorb outgoing terrestrial infrared radiation matching natural vibration frequencies of polar C-H bonds [1] This increases vibrational kinetic energy, which is re-emitted in all directions, trapping heat in the troposphere [1]	2
13(b)	Methane absorbs infrared radiation more strongly in atmospheric window wavelengths where CO2 does not absorb [1]	1
14	Correct Answer: C [1] Explanation: Products: 1 C8H18 + 2 C3H6 + 1 C2H4 = C(8 + 6 + 2) H(18 + 12 + 4) = C16H34 (hexadecane).	1
15(a)	Curly arrow from C=C double bond to H(delta+) of H-Br [1] Curly arrow showing heterolytic cleavage from H-Br bond to Br(delta-) [1] Correct structure of secondary carbocation intermediate: CH3-CH-CH3 [1] Curly arrow from lone pair on :Br to the positively charged carbon atom C [1]	4
15(b)	2-bromopropane [1]	1
15(c)	The major product forms via the secondary carbocation, CH3-CH-CH3, whereas the minor product forms via the primary carbocation, CH3-CH2-CH2 [1] The secondary carbocation has two electron-donating alkyl (methyl) groups attached to C, which exert a positive inductive effect [1] This disperses the positive charge more effectively, stabilizing the intermediate and lowering the activation energy [1]	3
16(a)	With cyclohexene: Orange-brown colour decolourises immediately / turns colourless [1] With cyclohexane: No visible change / solution remains orange-brown [1]	2
16(b)	Equation: CH2=CH2 + Br2 -> CH2Br-CH2Br [1] Type of mechanism: Electrophilic addition [1]	2
17(a)	The carbocation intermediate (or cyclic bromonium ion) is attacked by water molecules which are present in vast excess compared to bromide ions [1] Loss of a proton from the oxonium ion intermediate yields 2-bromoethanol: CH2BrCH2-OH2 -> CH2BrCH2OH + H [1]	2
17(b)	Chloride ions (:Cl) from the dissolved NaCl act as competing nucleophiles [1] They attack the positively charged carbon of the bromo-carbocation intermediate: CH2Br-CH2 + Cl -> CH2Br-CH2Cl [1]	2
18(a)	Reagents: Ethene and steam (H2O(g)) [1] Conditions: 300 °C and 60 - 70 atm [1] Catalyst: Concentrated phosphoric(V) acid, H3PO4 (absorbed on silica) [1]	3
18(b)	CH2=CH2(g) + H2O(g) -> CH3CH2OH(g or l) [1]	1
18(c)	Fermentation uses renewable plant biomass (sugar cane / maize) rather than non-renewable crude oil fractions / is closer to carbon-neutral [1]	1
19	Correct Answer: A [1] Explanation: Catalytic hydrogenation of C=C double bonds in polyunsaturated vegetable oils is carried out using a finely divided nickel catalyst at approximately 150 °C (or platinum at room temperature).	1
20(a)	Correct skeletal formula of 2-chloro-2-methylbutane [1] Name: 2-chloro-2-methylbutane [1]	2
20(b)	Electrophilic attack by H on C3 generates a tertiary (3°) carbocation, (CH3)2C-CH2CH3 [1] A tertiary carbocation is much more stable than the alternative secondary carbocation due to electron donation from three alkyl groups [1]	2
21(a)	The high electron density in the exposed pi bond repels the electron cloud of the approaching bromine molecule [1] This induces a temporary dipole, making the closer bromine atom partially positive (Br_delta+) and the further bromine atom partially negative (Br_delta-) [1]	2

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
21(b)	The Br-Br bond undergoes heterolytic fission, releasing a bromide ion (:Br) [1]	1
22	Correct Answer: B [1] Explanation: By Markovnikov's rule, H adds to C1 (which has 2 hydrogens), forming the more stable secondary carbocation CH ₃ -CH-CH ₃ . Addition of water and deprotonation yields propan-2-ol as the major product.	1
23(a)	CH ₂ =CH-CH(Br)-CH ₂ Br [1]	1
23(b)	Correct skeletal structure of 1,4-dibromobut-2-ene [1] Correctly showing the (E)-stereochemistry (Br-CH ₂ groups on opposite sides of the C=C double bond) [1]	2
24	Correct Answer: B [1] Explanation: Cyclohexene has one double bond. Electrophilic addition of HBr across the double bond yields bromocyclohexane (C ₆ H ₁₁ Br).	1
25(a)	Moles of H ₂ = 4.80 / 24.0 = 0.200 mol [1] Ratio: 0.200 mol H ₂ / 0.100 mol hydrocarbon = 2 moles of H ₂ per mole; therefore, 2 C=C double bonds (or 1 triple bond) [1]	2
25(b)	Buta-1,3-diene: CH ₂ =CH-CH=CH ₂ [1] Buta-1,2-diene: CH ₂ =C=CH-CH ₃ (or but-1-yne / but-2-yne) [1]	2
26(a)	Iodine atom (I has lower electronegativity than Cl, so it is polarised I _δ ⁺ - Cl _δ ⁻) [1]	1
26(b)	Major product: 2-chloro-1-iodopropane, CH ₃ -CH(Cl)-CH ₂ I [1] I adds to C1 to give more stable secondary carbocation CH ₃ -CH-CH ₂ I, followed by attack of Cl at C2 [1]	2
27	Correct Answer: C [1] Explanation: But-2-ene (CH ₃ -CH=CH-CH ₃) is completely symmetrical; addition of H to either C2 or C3 produces the identical carbocation CH ₃ -CH-CH ₂ CH ₃ and yields 2-bromobutane as the exclusive product.	1
28(a)	(i) =CH ₂ oxidises to carbon dioxide (CO ₂) and water (H ₂ O) [1] (ii) =CH-R oxidises to a carboxylic acid (R-COOH) [1] (iii) =C(R1)(R2) oxidises to a ketone (R1-CO-R2) [1]	3
28(b)	Symmetrical hex-3-ene: CH ₃ -CH ₂ -CH=CH-CH ₂ -CH ₃ [1] Name: Hex-3-ene [1]	2
28(c)	Structural formula: (CH ₃) ₂ C=CH ₂ (2-methylpropene) [1] Name: 2-methylpropene (or methylpropene) [1]	2
29(a)	Purple solution decolourises (or turns colourless / brown precipitate of MnO ₂ forms if alkaline/neutral) [1]	1
29(b)	CH ₃ -CH=CH ₂ + [O] + H ₂ O → CH ₃ -CH(OH)-CH ₂ OH [2] (1 mark for reactants, 1 mark for correct diol structure)	2
29(c)	Propane-1,2-diol [1]	1
30	Correct Answer: A [1] Explanation: Combining CH ₃ -C(=O)-CH ₂ CH ₃ and HOOC-CH ₃ gives CH ₃ -C(CH ₂ CH ₃)=CH-CH ₃ , which is 3-methylhex-2-ene.	1
31(a)	HOOC-CH ₂ -CH ₂ -CH ₂ -CH ₂ -COOH (or HOOC-(CH ₂) ₄ -COOH) [1]	1
31(b)	Hexanedioic acid (adipic acid) [1]	1
31(c)	The C=C double bond requires 4 equivalents of [O] (or 4/5 moles of MnO ₄) [1]	1
32(a)	Both molecules contain a C=C double bond, so both will rapidly decolourise bromine water from orange-brown to colourless [1]	1
32(b)	Heat each sample with hot concentrated acidified KMnO ₄ [1] Hex-1-ene has a terminal =CH ₂ group which oxidises to CO ₂ gas, producing visible effervescence (bubbles of gas that turn limewater milky) [1] Hex-3-ene has an internal double bond which oxidises quietly to propanoic acid with NO gas evolved / no effervescence [1]	3

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
33	Correct Answer: A [1] Explanation: The ring C=C has =C(CH ₃)- and =CH-. Cleavage converts =C(CH ₃)- into a methyl ketone (CH ₃ -CO-) and =CH- into a carboxylic acid (-COOH). The 5-carbon ring opens to yield 5-oxohexanoic acid.	1
34(a)	C=C alkene absorption at 1620 - 1680 cm ⁻¹ (or =C-H stretch at 3000-3100 cm ⁻¹) [1]	1
34(b)	Broad absorption at 3200 - 3650 cm ⁻¹ [1] O-H alcohol stretch (hydrogen bonded) [1]	2
35	Correct Answer: C [1] Explanation: Only terminal alkenes containing a =CH ₂ group produce CO ₂ on oxidative cleavage with hot concentrated KMnO ₄ . 3-methylbut-1-ene has a terminal double bond.	1
36(a)	Limonene has two C=C bonds, so 0.100 mol reacts with 0.200 mol Br ₂ [1] Mass of Br ₂ = 0.200 * 159.8 = 31.96 g [1] Volume = mass / density = 31.96 / 3.10 = 10.3 cm ³ [1]	3
36(b)	The exocyclic double bond has a terminal =CH ₂ group which is oxidised completely to CO ₂ and H ₂ O [1]	1
37(a)	Moles of CO ₂ = 1.20 / 24.0 = 0.0500 mol [1] Ratio: 0.0500 mol CO ₂ / 0.0500 mol alkene = 1 =CH ₂ group per molecule [1]	2
37(b)	Structure: (CH ₃) ₂ C=CH ₂ [1] Name: 2-methylpropene [1]	2
38	Correct Answer: B [1] Explanation: Cold dilute acidified KMnO ₄ oxidises the C=C double bond into a vicinal diol without cleaving the carbon-carbon sigma bond. But-2-ene forms butane-2,3-diol, CH ₃ CH(OH)CH(OH)CH ₃ .	1
39	Hexane: Remains purple / no reaction [1] Hex-1-ene: Purple solution decolourises immediately / turns colourless [1] Explanation: Hexane contains only strong, unreactive C-C and C-H sigma bonds; hex-1-ene contains an electron-rich, reactive pi bond susceptible to electrophilic attack by the oxidising agent [1]	3
40(a)	Repeat unit showing two backbone carbons with single bond and open continuation bonds: -[CH ₂ -CH(CH ₃)]- [1] Displayed with brackets and 'n' (or continuation bonds passing through brackets) [1]	2
40(b)	Displayed formula of chloroethene: CH ₂ =CH-Cl (with all bonds shown) [1] Name: Chloroethene (or vinyl chloride) [1]	2
40(c)	The polymer backbone consists exclusively of strong, non-polar C-C and C-H single bonds with high bond enthalpies [1] There are no polar bonds, delta+ sites, or functional groups that bacteria or environmental enzymes can attack or hydrolyse [1]	2
41(a)	Landfill sites occupy large areas of land and take up valuable space / visual pollution [1] Plastics persist for hundreds of years without decomposing; can break into harmful microplastics / harm wildlife [1]	2
41(b)	Toxic gas: Hydrogen chloride, HCl (or chlorine, Cl ₂ / chlorinated dioxins) [1] Removal: Scrubber using basic calcium oxide / calcium carbonate / sodium hydroxide to neutralise the acidic gas [1]	2
41(c)	Thermal cracking of polymer waste back into original monomers or hydrocarbon petrochemical feedstock to synthesize new chemicals [1]	1
42	Correct Answer: B [1] Explanation: The monomer for poly(tetrafluoroethene) is tetrafluoroethene, CF ₂ =CF ₂ . During addition polymerisation, the C=C double bond opens to form the repeat unit -[CF ₂ -CF ₂]-.	1
43(a)	Displayed formula showing CH ₂ =C(CH ₃)COOCH ₃ with all bonds shown [2] (1 mark if partially correct)	2
43(b)	Methyl 2-methylpropenoate (methyl methacrylate) [1]	1
43(c)	Perspex / Plexiglas / PMMA [1]	1
44	Correct Answer: B [1] Explanation: But-2-ene is CH ₃ -CH=CH-CH ₃ . The polymer backbone is formed by the two alkene carbons (C ₂ and C ₃), each bearing one H and one -CH ₃ group: -[CH(CH ₃)-CH(CH ₃)]-.	1

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
45(a)	Structure showing: $-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}(\text{CH}_3)-$ with open continuation bonds [2] (1 mark if partially correct)	2
45(b)	Moles of propene = $42.0 / 42.0 = 1.00 \text{ mol}$ [1] Alternating copolymer requires 1:1 mole ratio; Mass of ethene = $1.00 \text{ mol} \times 28.0 \text{ g mol}^{-1} = 28.0 \text{ g}$ [1]	2
46	Correct Answer: B [1] Explanation: Polyesters and polyamides have polar linkages ($-\text{COO}-$, $-\text{CONH}-$) susceptible to acid or alkaline hydrolysis by environmental moisture and microorganisms, whereas polyalkenes have inert non-polar C-C backbones.	1
47(a)	$\text{CH}_2=\text{C}(\text{CH}_3)-\text{CH}=\text{CH}_2$ [1]	1
47(b)	$-\text{CH}_2-\text{C}(\text{CH}_3)=\text{CH}-\text{CH}_2-$ [2] (1 mark for correct backbone, 1 mark for double bond at C2-C3)	2
47(c)	The cis-isomer has kinky, disordered chains that cannot pack closely together, resulting in weak intermolecular forces that allow chains to stretch and recoil [1] The trans-isomer has regular, straight chains that pack closely into a crystalline lattice with stronger intermolecular forces, making it rigid [1]	2
48	Correct Answer: B [1] Explanation: Polymers are derived from petroleum and have high calorific values (comparable to fuel oil). Modern incinerators burn them to produce high-pressure steam for electricity turbines, recovering energy.	1
49(a)	Different polymers have different melting temperatures and do not mix well / are mutually immiscible at molecular level [1] Melting an unsorted blend produces a brittle, mechanically defective recycled plastic of very low quality and value [1]	2
49(b)	Flotation / density separation in liquid media (or near-infrared NIR optical spectroscopy sorting) [1]	1
50(a)	Hydrocarbon A is but-2-ene, $\text{CH}_3-\text{CH}=\text{CH}-\text{CH}_3$ [1] Type of stereoisomerism: Geometric (cis/trans or E/Z) isomerism [1]	2
50(b)	Displayed formula of 2,3-dibromobutane: $\text{CH}_3-\text{CH}(\text{Br})-\text{CH}(\text{Br})-\text{CH}_3$ [1] Name: 2,3-dibromobutane [1]	2
50(c)	Compound C is ethanoic acid, CH_3COOH [1] Equation: $2\text{CH}_3\text{COOH} + \text{Na}_2\text{CO}_3 \rightarrow 2\text{CH}_3\text{COONa} + \text{H}_2\text{O} + \text{CO}_2$ [1]	2
50(d)	Compound D is butane, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$ [1] Two structural isomers: 1-chlorobutane and 2-chlorobutane [1]	2